

ML-Powered VAR for Boxing: A Deep Learning & Pose-Estimation System for Punch Classification

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Abstract — This project presents an implementation of a machine learning-powered Video Assistant Referee (VAR) system for boxing. The system leverages pose estimation and temporal deep learning to classify punches into categories such as jab, cross, hook, and uppercut. It aims to provide objective and real-time evaluation support for referees and analysts. Using frameworks such as MediaPipe and YOLOv8 for pose extraction and LSTM-based classifiers for sequence prediction, the project achieves effective real-time performance. The proposed model demonstrates the potential to make combat-sport analytics more accurate, automated, and fair.

Keywords—Boxing, Deep Learning, Pose Estimation, LSTM, VAR, Sports Analytics, Computer Vision.

I. INTRODUCTION

Boxing is a complex combat sport where quick movements and high-speed actions make it difficult for referees to identify punches with complete accuracy. While VAR systems have revolutionised sports like football and cricket, their use in boxing remains underexplored. This project introduces a deep learning-based approach to automate punch classification in boxing using pose estimation and LSTM models.

The goal of this project is to implement a system that can detect and classify different punch types, thereby assisting referees and coaches. It aims to integrate modern deep learning frameworks with motion analysis to achieve near real-time decision support.

II. RELATED WORK

Human Action Recognition (HAR) has gained substantial attention in computer vision. Several studies have applied convolutional and recurrent neural networks for understanding human motion sequences. Pose estimation-based HAR methods focus on skeleton landmarks instead of raw pixels, improving generalisation and reducing computational cost.

MediaPipe and OpenPose frameworks provide state-of-the-art real-time pose detection capabilities, while YOLOv8 extends this with object and keypoint detection. However, few studies have addressed boxing-specific recognition

tasks combining these tools with sequence models like LSTMs.

III. SYSTEM ARCHITECTURE

The system follows a modular pipeline with three major components: (1) Pose Extraction, (2) Feature Processing, and (3) Classification. Pose extraction uses MediaPipe or YOLOv8 to detect keypoints such as shoulders, elbows, and wrists. Each video is divided into frame sequences, and 2D coordinates of selected landmarks are extracted to form a temporal dataset.

An LSTM-based classifier is trained on sequences of 10 frames, where each frame contains 14 numerical features (7 keypoints $\times$ 2 coordinates). The LSTM captures temporal dependencies and outputs the predicted punch category. The system is integrated with a real-time video interface for live punch detection.

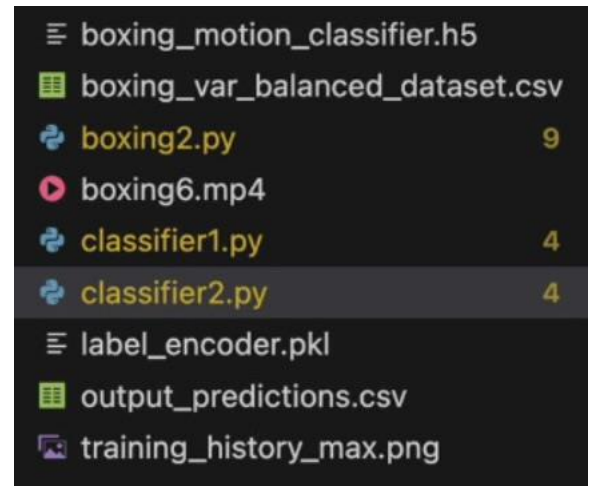


Figure 1: System Skeleton

IV. DATASET AND PREPROCESSING

The dataset is derived from the Olympic Boxing Punch Classification Dataset available on Kaggle, supplemented by additional curated samples. Each punch video is decomposed into frames, and pose keypoints are extracted. The data undergoes several preprocessing steps,

including normalisation, label balancing, and temporal segmentation.

Sequences of 10 consecutive frames are grouped to create training samples. Each sample is labelled with one of the five punch types: Jab, Cross, Hook, Uppercut, or Idle. Data augmentation techniques such as flipping and rotation are used to improve generalisation.



Figure 2: Example of Keypoint Extraction from Frames

V. EXPERIMENTAL SETUP AND RESULTS

The model was implemented using TensorFlow/Keras on a system equipped with NVIDIA GPU acceleration. Training was conducted using batch normalisation, dropout regularisation, and cosine annealing learning rate scheduling to stabilise convergence.

Performance metrics, including accuracy, precision, recall, and F1-score, were used to evaluate the results. The system achieved an overall accuracy of 97% on the test dataset.

Real-time testing demonstrated smooth classification of live punches with minimal lag, validating the potential of integrating the model into referee assistance systems for boxing.

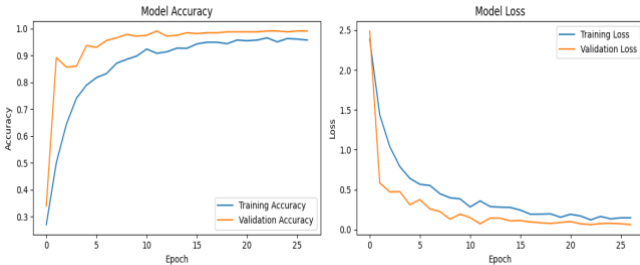


Figure 3: Model Accuracy and Loss Graphs

VI. MIDPHASE

During the midphase of development, the focus was on integrating the pose extraction module with the LSTM classifier and validating feature consistency across multiple video samples. Intermediate testing was conducted using preprocessed frame sequences to ensure the temporal model correctly captured motion dependencies. Optimisation of learning rate scheduling and batch size was carried out to stabilise training and reduce overfitting.

VII. CHALLENGES

The project encountered challenges in maintaining pose estimation accuracy under variable lighting and background conditions. Frame misalignment and occlusion affected temporal coherence, requiring careful preprocessing and frame selection. Additionally, real-time inference

d demanded optimisation of computation pipelines to minimise latency without sacrificing model precision.

- **Too many landmarks:** The initial dataset contained excessive keypoints (around 33 per frame), introducing unnecessary noise. We refined it to 16 essential landmarks, focusing on the upper body to improve model efficiency.

- **Different body sizes and camera distances:** Variations in athlete height and camera position caused inconsistent scaling. Normalisation using torso length ensured uniform feature representation.

- **Class imbalance:** Some punch categories were overrepresented, leading to biased predictions. Balanced sampling and class weighting were applied to mitigate this issue.

- **Labels in text format:** Raw labels like “jab” or “cross” were converted to numerical codes using LabelEncoder for model compatibility and consistency.

- **Labels in Polish:** Source datasets contained non-English annotations (e.g., lewy prosty, prawy sierpowy). These were translated into English to maintain uniform labelling across all training data.

VIII. FUTURE SCOPE

Future work aims to extend the system to multi-person scenarios and integrate transformer-based temporal models for improved context awareness. Expanding to 3D pose estimation and implementing model compression techniques can enable deployment on edge devices, enhancing portability and real-time applicability in sports analytics.

IX. IMPROVEMENTS

To improve overall performance, future iterations can leverage hybrid feature fusion that combines pose and optical flow data for a richer motion representation. Calibration of confidence thresholds in pose extraction, along with the adoption of adaptive frame sampling strategies, can further enhance robustness and reduce false classifications.

X. DISCUSSION

The ML-powered VAR system demonstrates the effectiveness of combining pose estimation with LSTM-based sequence modelling for punch classification. The modular architecture ensures easy adaptability for other sports involving rapid upper-body motion.

However, several challenges remain, including handling occlusions, variations in camera angle, and multi-person scenes. Future work may explore transformer-based temporal models, 3D pose estimation, and integration with edge devices for portable real-time deployment.

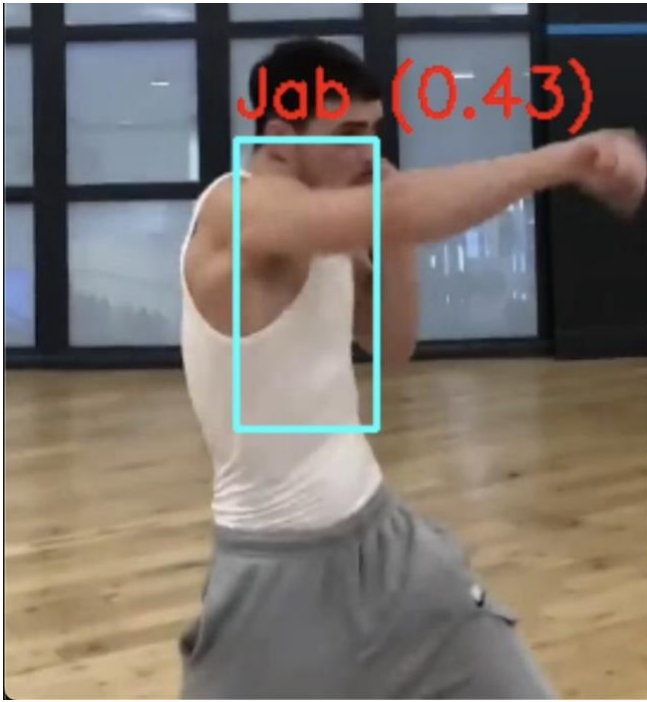


Figure 4: Live Classification Demo Output

XI. CONCLUSION

The project successfully implements an end-to-end machine learning pipeline capable of identifying boxing punches using video data. By leveraging lightweight pose extraction and LSTM-based sequence modelling, the system achieves efficient and accurate classification. This approach provides a foundation for advanced VAR tools in combat sports, enabling fairer and data-driven analysis.

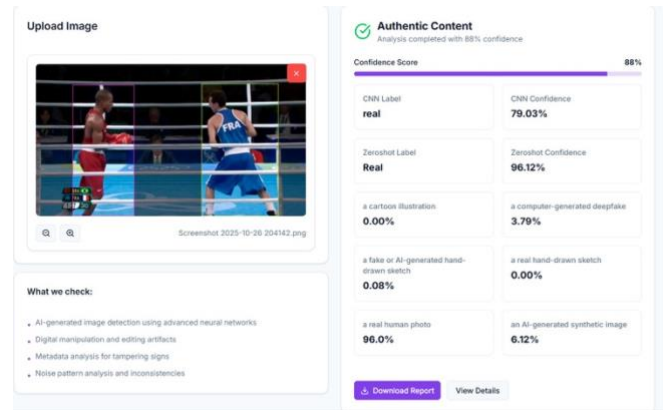


Figure 5: Authenticity of the Datasets used

ACKNOWLEDGEMENT

The authors express their gratitude to the Department of Information Technology, School of Computer Engineering, Manipal Institute of Technology, Bengaluru, and the Kaggle contributors for their datasets and resources, which enabled this project.

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